

JOCELYN XU

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EDUCATION

University of Waterloo

Computer Science, Honours, Co-op Program

Sept 2024

Waterloo, ON

- CAV: 95.3

EXPERIENCE

Bank of Montreal (BMO) Software Developer | *Angular, AWS, TypeScript, Figma*

May 2025 – Dec 2025

Help Center AI Chatbot

- Architected a **serverless RAG chatbot** using **AWS Lambda** and **Amazon Bedrock**, implementing **WebSockets** for real-time response streaming to drive a **30%** increase in internal company queries
- Engineered a **multi-agent routing** system using a supervisor-collaborator architecture to classify query intent, reducing latency by **35%** through optimized agent selection and **LRU caching** of frequent queries
- Eliminated **20+ hours** of monthly manual data entry by building a scheduled Puppeteer scraper that extracts, chunks, and embeds 125+ Online Banking documents into S3 for real-time RAG retrieval

Lexicon Design System (LDS)

- Scaled **LDS** adoption to **300+** developers by engineering **15+ framework-agnostic** web components to replace legacy Angular components, ensuring universal **WCAG 2.2 AA** compliance across all platforms
- Improved code coverage to **80%+** by authoring **250+** Jest and Playwright tests, resolving **360+** bugs in shared components used by **300+** developers, preventing repeated defects and reducing QA effort across teams
- Streamlined delivery for BMO products serving **60,000+** clients by orchestrating bi-weekly releases with semantic versioning and publishing bundled builds to **JFrog Artifactory** as internal NPM packages

Data Science Club VP Development | *Typescript, React, Node.js, PostgreSQL, Shadcn/ui*

Dec 2024 – Apr 2026

- Led development of a full-stack AI hackathon website serving **1,100+** users, building a check-in system with HMAC-signed URLs for QR codes and NFC tags, reducing manual processing time from **60s** to under **5s**
- Automated infrastructure for development teams by engineering **GitHub Actions CI/CD workflows** with Infsical secrets management and **Discord webhooks**, reducing manual project management overhead from **5+ minutes to seconds** via automated deployment previews and issue tracking
- Refactored backend to a **Controller-Service-Repository** architecture during a **MongoDB to Supabase migration**, implementing relational schemas and **Row-Level Security** to boost data integrity and query efficiency by **40%**

PROJECTS

Catan Solver | *PyTorch, OpenCV, YOLOv8, Multimodal AI Agents, Statistic Models*

Jan 2026

- Engineered a hybrid multi-threaded AI agent capable of playing competitive Catan, integrating computer vision, probabilistic reasoning, and LLM-based decision making
- Fine-tuned a **YOLOv8m-seg** model on **500+** annotated screenshots to detect hex tiles and ports, achieving **98.5%** mAP with sub-**100ms** inference latency
- Achieved a **45%** win rate by designing a Bayesian uncertainty engine tracking opponent hands at **92%** accuracy and Monte Carlo simulations evaluating **10,000+** board states per turn

Fashionkilla | *React, Tailwind, MongoDB, FastAPI, LLM, Sentence Transformers*

Oct 2025

- Recognized with **Best Use of MongoDB** and **Best UI/UX** at TechNova 2025 Hackathon
- Engineered an AI fashion recommendation engine using Sentence Transformers for vector embeddings, implementing cosine similarity for semantic retrieval and **FastAPI** for CRUD operations
- Architected a data pipeline using **Google Gemini** for image context extraction, optimizing latency by caching **vector embeddings** and image encoded data in MongoDB to eliminate redundant API data calls

ResuMix | *Python, Flask, React, Vite, NeonDB, Sentence Transformers, ReportLab*

Aug 2025

- Developed a resume **ATS optimization** tool using Sentence Transformers and DP algorithms, outperforming legacy GPT-wrappers with a **30%** accuracy gain and **2000+ms** reduction in inference latency
- Architected a **NeonDB** backend with connection pooling and **batch commands** to handle **high-concurrency** traffic, improving query performance by **45%** during PDF generation

SKILLS

- **Languages:** Python, C++, JavaScript, TypeScript, SQL, Java, Bash, HTML, CSS
- **Frameworks:** React, Next.js, Angular, Flask, FastAPI, Express.js, Tailwind
- **Tools:** AWS, PostgreSQL, Docker, Node.js, Postman, Figma